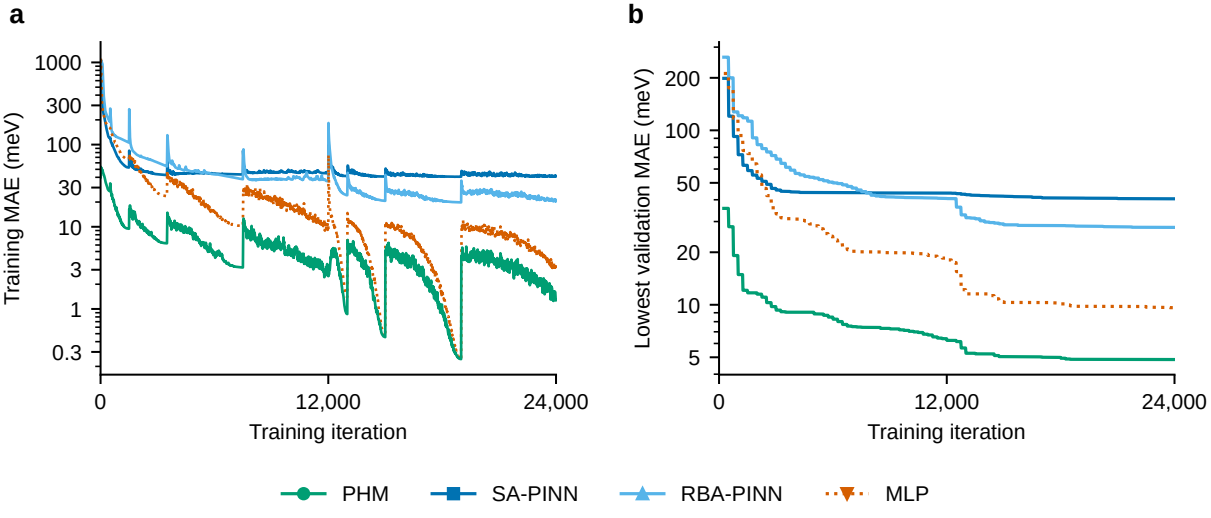

a
b
Training MAE (meV)
1000
300
100
30
10
3
1
0.3
0
12,000
24,000
Training iteration
Lowest validation MAE (meV)
200
100
50
20
10
5
0
12,000
24,000
Training iteration
PHM
SA-PINN
RBA-PINN
MLP